Experiment No:9
Experiment to understand Matplotlib library use cases in Data Science through visualization

AIM:

To understand Matplotlib library use cases in Data Science through visualization

DESCRIPTION:

Use Iris data set to understand the Matplot lib library

CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline

# Load the dataset
data = pd.read_csv('/content/Iris.csv')

# Display the dataset
data

# Display dataset information
data.info()

# Statistical summary
data.describe()

# Count the number of flowers in each species
data.value_counts('Species')
```

```
# Count plot
sns.countplot(x='Species', data=data)
plt.show()

# Convert categorical data into dummy variables
dummies = pd.get_dummies(data['Species'])

# Combine dummy variables with numerical columns
FinalDataset = pd.concat(
    [dummies, data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

# Display the final dataset
FinalDataset.head()

# Scatter plot: Sepal Length vs Sepal Width
sns.scatterplot(
    x='SepalLengthCm',
    y='SepalWidthCm',
    hue='Species',
    data=data
)
plt.show()

# Scatter plot: Petal Length vs Petal Width
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='Species',
    data=data
)
```

```
plt.show()

# Pair plot
sns.pairplot(data, hue='Species', height=3)
plt.show()

# Histogram: Petal Length
sns.FacetGrid(data, hue='Species', height=5).map(
    sns.histplot, 'PetalLengthCm'
).add_legend()
plt.show()

# Histogram: Petal Width
sns.FacetGrid(data, hue='Species', height=5).map(
    sns.histplot, 'PetalWidthCm'
).add_legend()
plt.show()

# Histogram: Sepal Length
sns.FacetGrid(data, hue='Species', height=5).map(
    sns.histplot, 'SepalLengthCm'
).add_legend()
plt.show()

# Histogram: Sepal Width
sns.FacetGrid(data, hue='Species', height=5).map(
    sns.histplot, 'SepalWidthCm'
).add_legend()
plt.show()
```

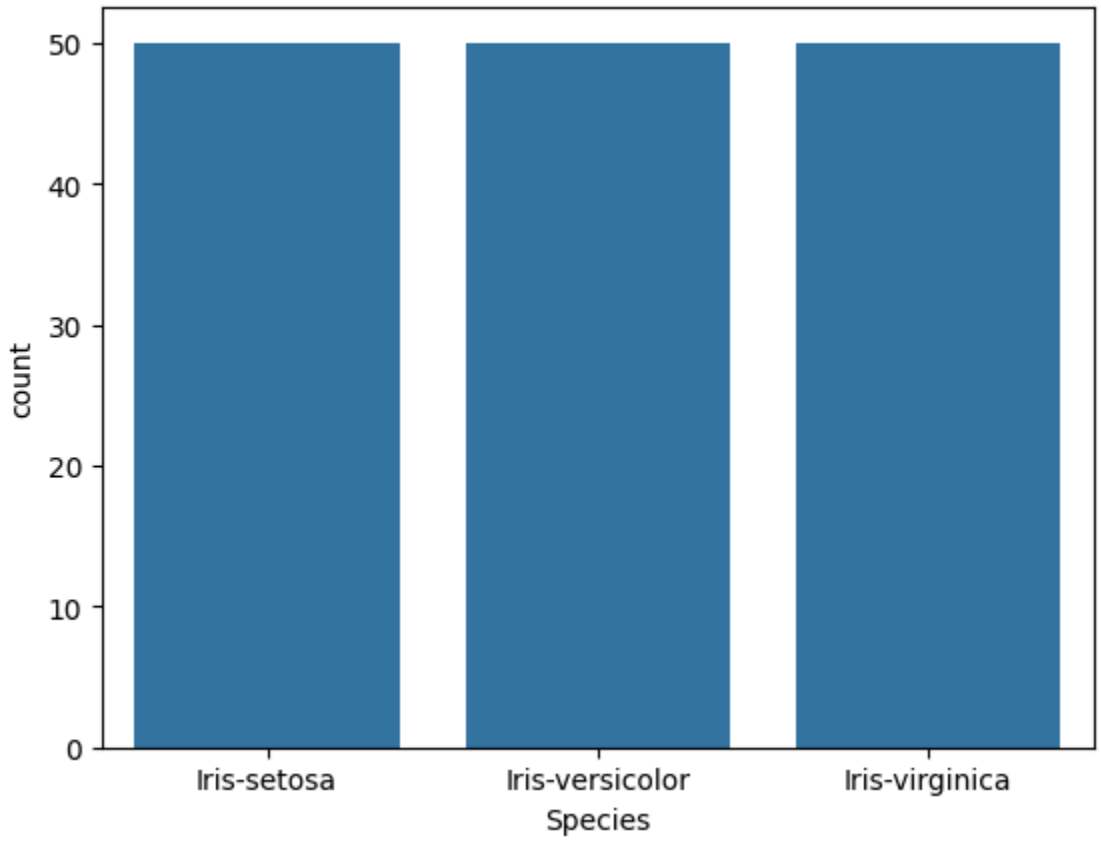
OUTPUT:

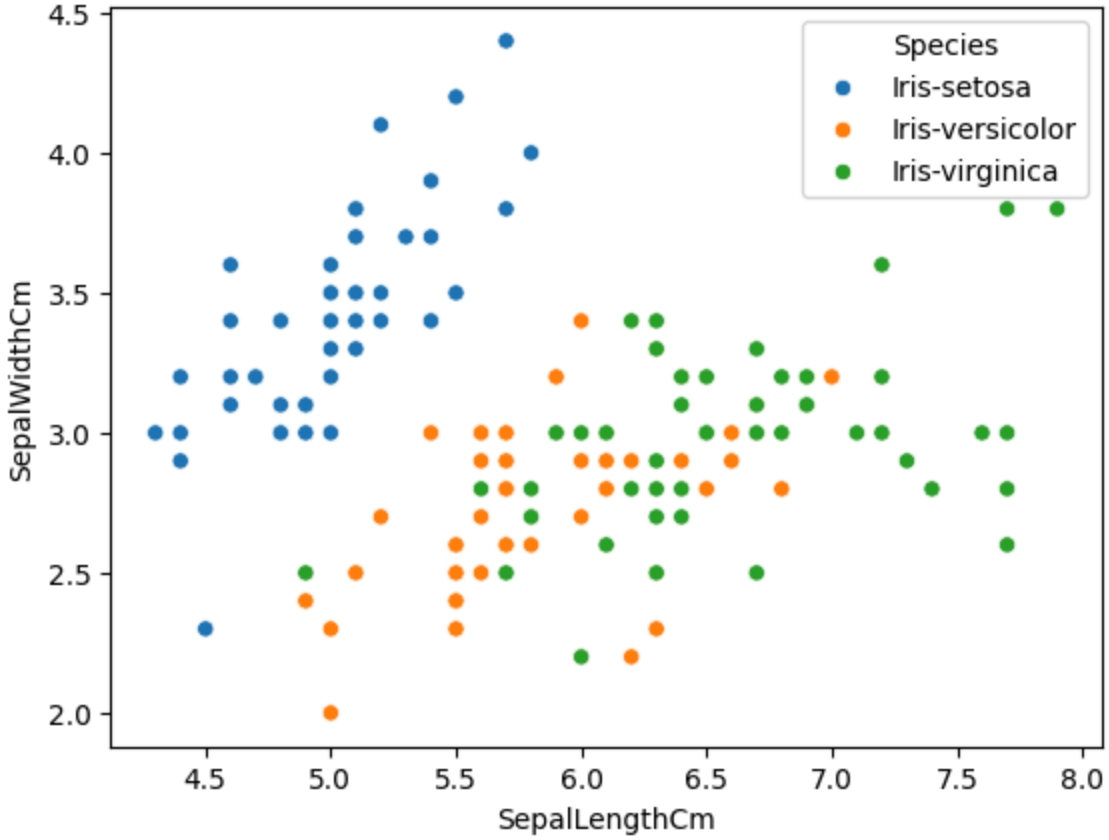
```
<class 'pandas.core.frame.DataFrame'>
```

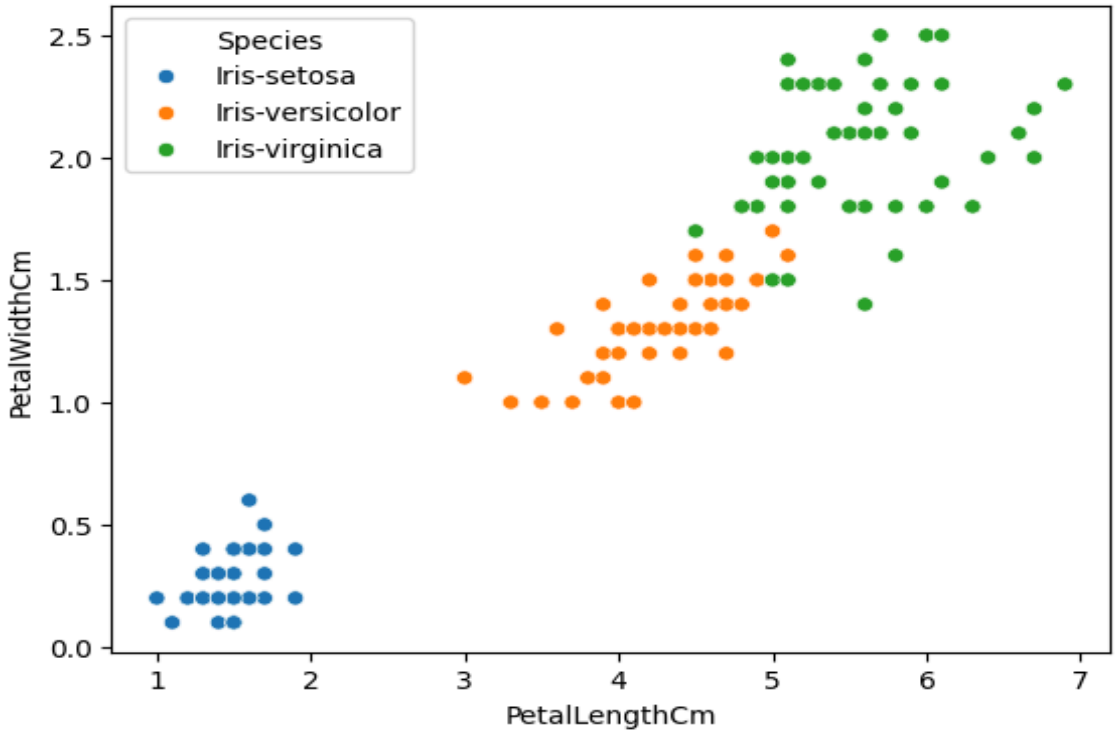
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):

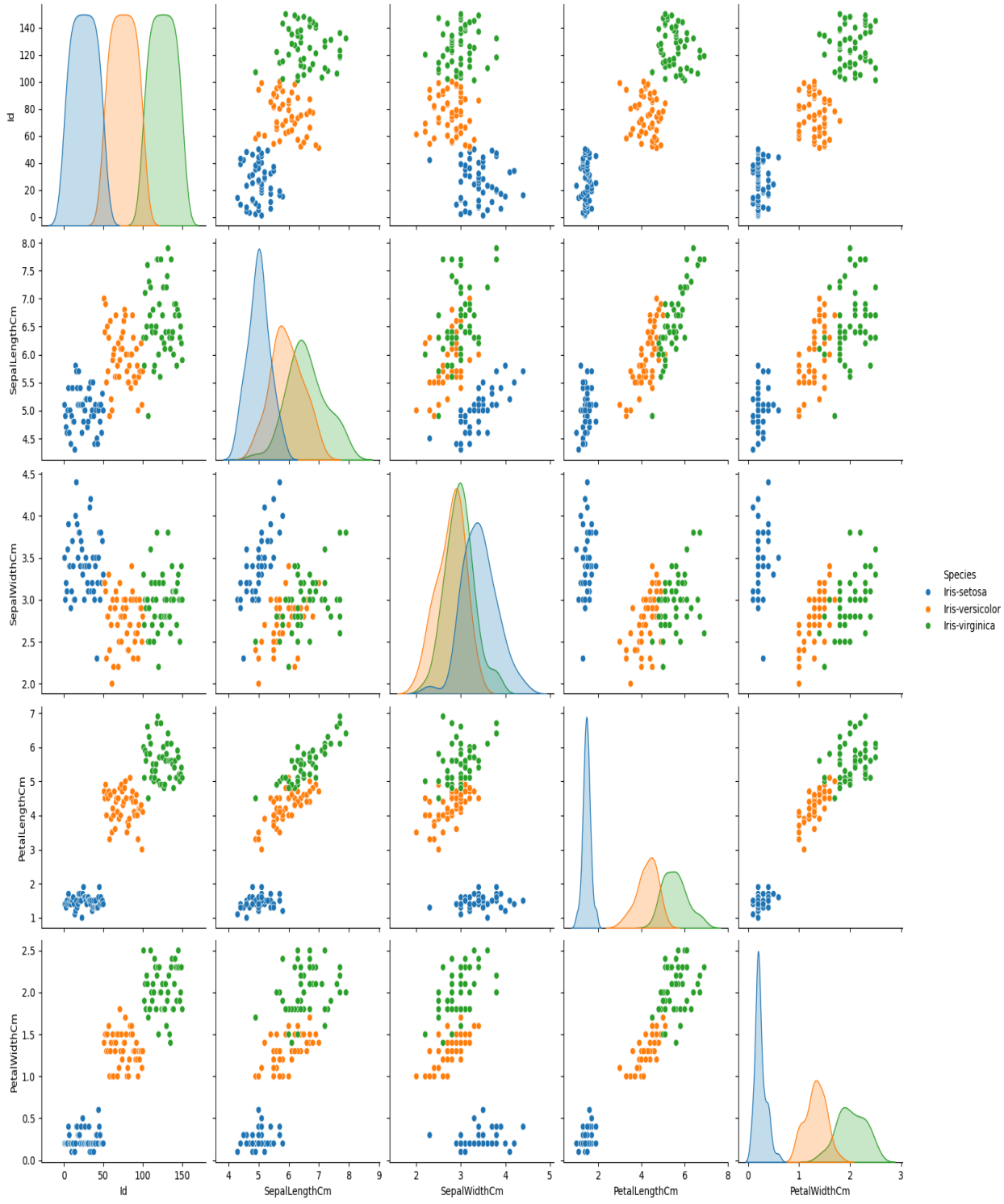
#	Column	Non-Null Count	Dtype
0	Id	150 non-null	int64
1	SepalLengthCm	150 non-null	float64
2	SepalWidthCm	150 non-null	float64
3	PetalLengthCm	150 non-null	float64
4	PetalWidthCm	150 non-null	float64
5	Species	150 non-null	object

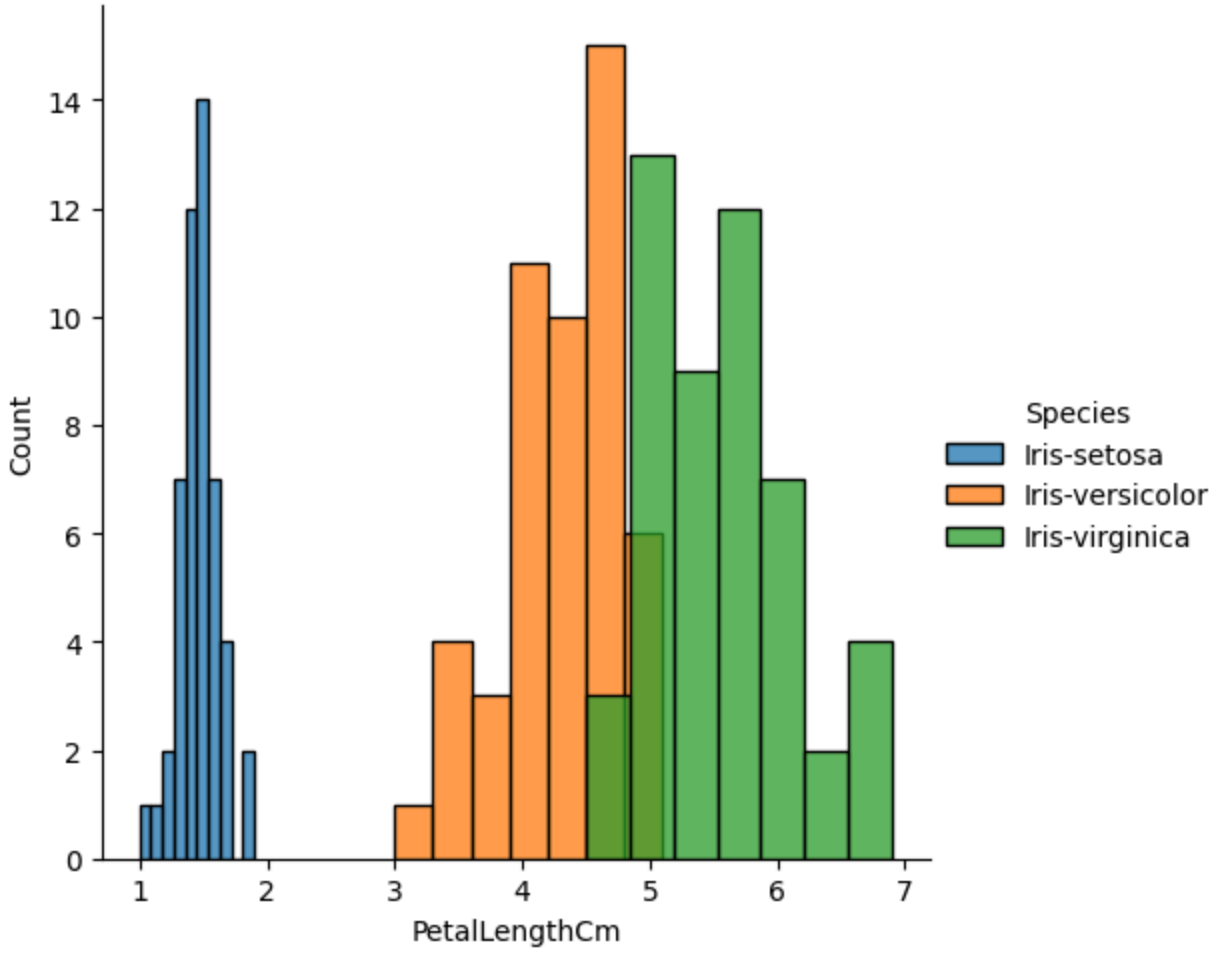
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB

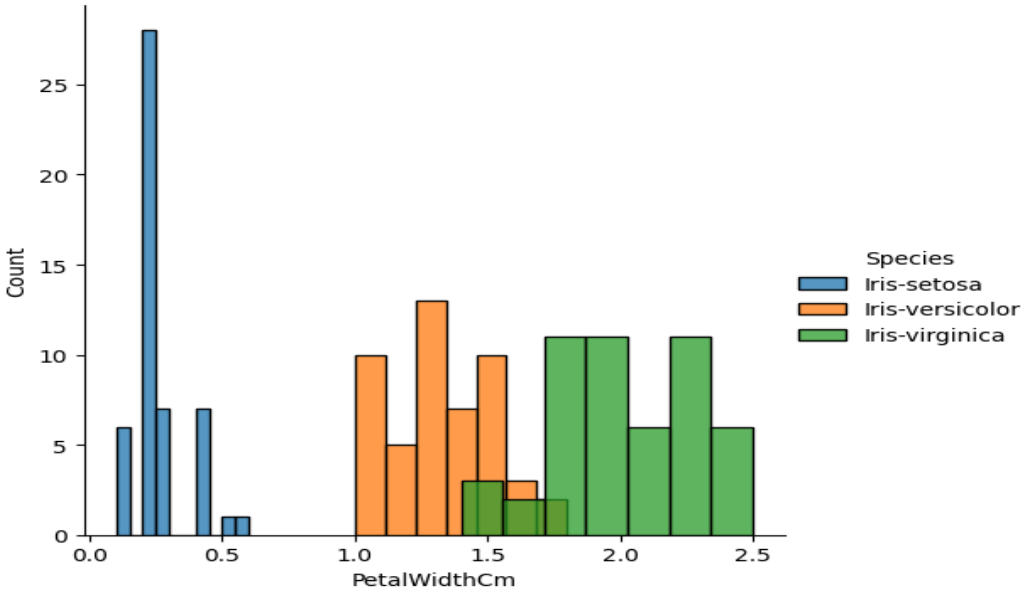


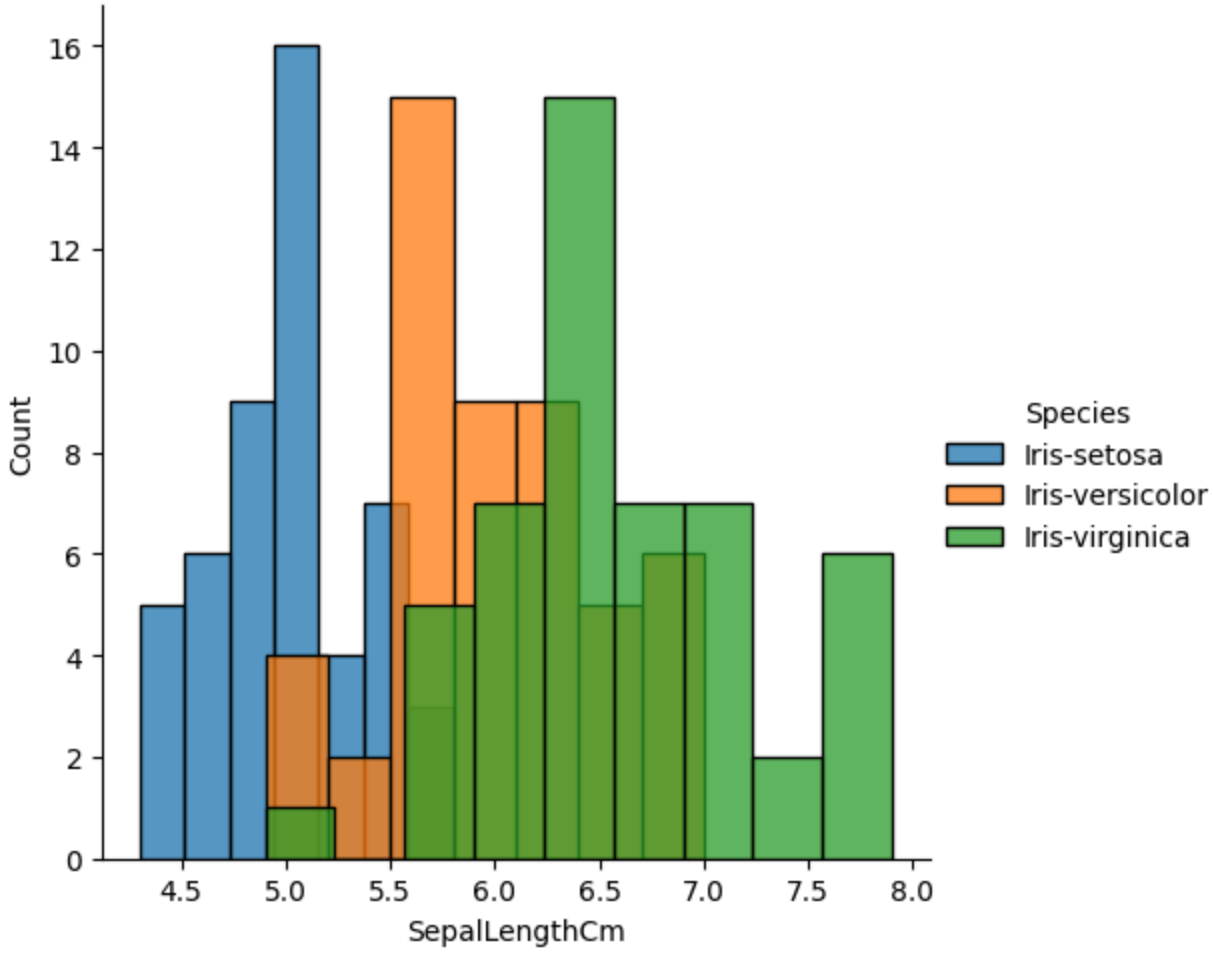

Species
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0

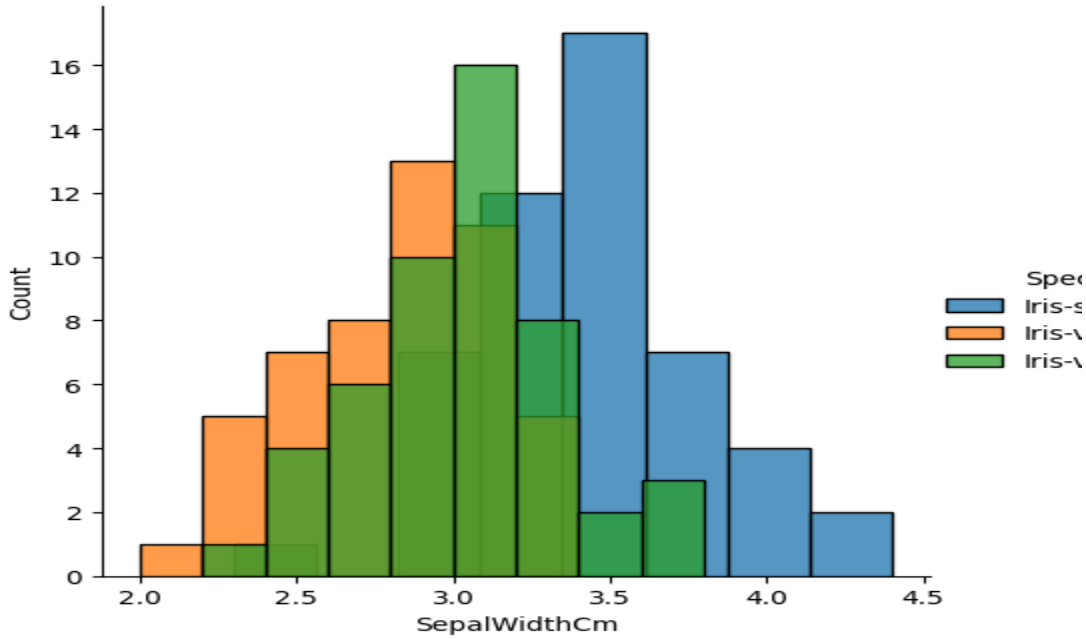

Species
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
Species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
Species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
Species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
Species
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalWidthCm

RESULT:

Thus, the Matplotlib library was successfully studied and implemented using the Iris dataset for data visualization. The dataset was analyzed using statistical summaries, count plots, scatter plots, pair plots, and histograms to visualize the distribution of flower features and compare the three Iris species.